

```
In [1]: import pandas as pd
```

```
In [2]: df=pd.read_csv("london_merged.csv")
```

```
In [3]: df
```

```
Out[3]:
```

	timestamp	cnt	t1	t2	hum	wind_speed	weather_code	is_holiday	is_weekend	season
0	2015-01-04 00:00:00	182	3.0	2.0	93.0	6.0	3.0	0.0	1.0	3.0
1	2015-01-04 01:00:00	138	3.0	2.5	93.0	5.0	1.0	0.0	1.0	3.0
2	2015-01-04 02:00:00	134	2.5	2.5	96.5	0.0	1.0	0.0	1.0	3.0
3	2015-01-04 03:00:00	72	2.0	2.0	100.0	0.0	1.0	0.0	1.0	3.0
4	2015-01-04 04:00:00	47	2.0	0.0	93.0	6.5	1.0	0.0	1.0	3.0
...	...	...	...	...	...	...	...	...	...	...
17409	2017-01-03 19:00:00	1042	5.0	1.0	81.0	19.0	3.0	0.0	0.0	3.0
17410	2017-01-03 20:00:00	541	5.0	1.0	81.0	21.0	4.0	0.0	0.0	3.0
17411	2017-01-03 21:00:00	337	5.5	1.5	78.5	24.0	4.0	0.0	0.0	3.0
17412	2017-01-03 22:00:00	224	5.5	1.5	76.0	23.0	4.0	0.0	0.0	3.0
17413	2017-01-03 23:00:00	139	5.0	1.0	76.0	22.0	2.0	0.0	0.0	3.0

17414 rows × 10 columns

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17414 entries, 0 to 17413
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   timestamp       17414 non-null  object
1   cnt              17414 non-null  int64
2   t1              17414 non-null  float64
3   t2              17414 non-null  float64
4   hum             17414 non-null  float64
5   wind_speed      17414 non-null  float64
6   weather_code    17414 non-null  float64
7   is_holiday      17414 non-null  float64
8   is_weekend      17414 non-null  float64
9   season          17414 non-null  float64
dtypes: float64(8), int64(1), object(1)
memory usage: 1.3+ MB
```

```
In [5]: df.shape
```

```
Out[5]: (17414, 10)
```

```
In [7]: df.isnull().sum()
```

```
Out[7]: timestamp    0
cnt                  0
t1                   0
t2                   0
hum                  0
wind_speed           0
weather_code         0
is_holiday           0
is_weekend           0
season               0
dtype: int64
```

```
In [8]: df['weather_code'].value_counts()
```

```
Out[8]: 1.0      6150
        2.0      4034
        3.0      3551
        7.0      2141
        4.0      1464
        26.0       60
        10.0       14
        Name: weather_code, dtype: int64
```

```
In [9]: df['season'].value_counts()
```

```
Out[9]: 0.0      4394
        1.0      4387
        3.0      4330
        2.0      4303
        Name: season, dtype: int64
```

```
In [10]: df.head()
```

```
Out[10]:
```

	timestamp	cnt	t1	t2	hum	wind_speed	weather_code	is_holiday	is_weekend	season
0	2015-01-04 00:00:00	182	3.0	2.0	93.0	6.0	3.0	0.0	1.0	3.0
1	2015-01-04 01:00:00	138	3.0	2.5	93.0	5.0	1.0	0.0	1.0	3.0
2	2015-01-04 02:00:00	134	2.5	2.5	96.5	0.0	1.0	0.0	1.0	3.0
3	2015-01-04 03:00:00	72	2.0	2.0	100.0	0.0	1.0	0.0	1.0	3.0
4	2015-01-04 04:00:00	47	2.0	0.0	93.0	6.5	1.0	0.0	1.0	3.0

```
In [11]: df.rename(columns={'timestamp':'time','cnt':'count','t1':'temp_real_C','t2':'temp_feel_C','hum':'humidity_percent','wind_speed':'wind_kph','weather_code':'weather'},inplace=True)
```

In [12]: df

Out[12]:

	time	count	temp_real_C	temp_feel_C	humidity_percent	wind_kph	weather	is_holiday	is_weekend	season
0	2015-01-04 00:00:00	182	3.0	2.0	93.0	6.0	3.0	0.0	1.0	3.0
1	2015-01-04 01:00:00	138	3.0	2.5	93.0	5.0	1.0	0.0	1.0	3.0
2	2015-01-04 02:00:00	134	2.5	2.5	96.5	0.0	1.0	0.0	1.0	3.0
3	2015-01-04 03:00:00	72	2.0	2.0	100.0	0.0	1.0	0.0	1.0	3.0
4	2015-01-04 04:00:00	47	2.0	0.0	93.0	6.5	1.0	0.0	1.0	3.0
...	...	...	...	...	...	...	...	...	...	...
17409	2017-01-03 19:00:00	1042	5.0	1.0	81.0	19.0	3.0	0.0	0.0	3.0
17410	2017-01-03 20:00:00	541	5.0	1.0	81.0	21.0	4.0	0.0	0.0	3.0
17411	2017-01-03 21:00:00	337	5.5	1.5	78.5	24.0	4.0	0.0	0.0	3.0
17412	2017-01-03 22:00:00	224	5.5	1.5	76.0	23.0	4.0	0.0	0.0	3.0
17413	2017-01-03 23:00:00	139	5.0	1.0	76.0	22.0	2.0	0.0	0.0	3.0

17414 rows × 10 columns

```
In [13]: #mapping the season and weather
season_dict={0.0:'Spring',
             1.0:'Summer',
             2.0:'Autumn',
             3.0:'Winter'}
weather_dict={1.0:'Clear',
              2.0:'Scattered Clouds',
              3.0:'Broken Clouds',
              4.0:'Cloudy',
              7.0:'Rain',
              10.0:'Rain with thnderstorm',
              26.0:'Snowfall'}
```

```
In [15]: df['season']=df['season'].map(season_dict)
```

```
In [16]: df
```

```
Out[16]:
```

	time	count	temp_real_C	temp_feel_C	humidity_percent	wind_kph	weather	is_holiday	is_weekend	season
0	2015-01-04 00:00:00	182	3.0	2.0	93.0	6.0	3.0	0.0	1.0	Winter
1	2015-01-04 01:00:00	138	3.0	2.5	93.0	5.0	1.0	0.0	1.0	Winter
2	2015-01-04 02:00:00	134	2.5	2.5	96.5	0.0	1.0	0.0	1.0	Winter
3	2015-01-04 03:00:00	72	2.0	2.0	100.0	0.0	1.0	0.0	1.0	Winter
4	2015-01-04 04:00:00	47	2.0	0.0	93.0	6.5	1.0	0.0	1.0	Winter
...	...	...	...	...	...	...	...	...	...	...
17409	2017-01-03 19:00:00	1042	5.0	1.0	81.0	19.0	3.0	0.0	0.0	Winter
17410	2017-01-03 20:00:00	541	5.0	1.0	81.0	21.0	4.0	0.0	0.0	Winter
17411	2017-01-03 21:00:00	337	5.5	1.5	78.5	24.0	4.0	0.0	0.0	Winter
17412	2017-01-03 22:00:00	224	5.5	1.5	76.0	23.0	4.0	0.0	0.0	Winter
17413	2017-01-03 23:00:00	139	5.0	1.0	76.0	22.0	2.0	0.0	0.0	Winter

17414 rows × 10 columns

```
In [18]: df['weather']=df['weather'].map(weather_dict)
```

In [19]: df

Out[19]:

	time	count	temp_real_C	temp_feel_C	humidity_percent	wind_kph	weather	is_holiday	is_weekend	season
0	2015-01-04 00:00:00	182	3.0	2.0	93.0	6.0	Broken Clouds	0.0	1.0	Winter
1	2015-01-04 01:00:00	138	3.0	2.5	93.0	5.0	Clear	0.0	1.0	Winter
2	2015-01-04 02:00:00	134	2.5	2.5	96.5	0.0	Clear	0.0	1.0	Winter
3	2015-01-04 03:00:00	72	2.0	2.0	100.0	0.0	Clear	0.0	1.0	Winter
4	2015-01-04 04:00:00	47	2.0	0.0	93.0	6.5	Clear	0.0	1.0	Winter
...	...	...	...	...	...	...	...	...	...	...
17409	2017-01-03 19:00:00	1042	5.0	1.0	81.0	19.0	Broken Clouds	0.0	0.0	Winter
17410	2017-01-03 20:00:00	541	5.0	1.0	81.0	21.0	Cloudy	0.0	0.0	Winter
17411	2017-01-03 21:00:00	337	5.5	1.5	78.5	24.0	Cloudy	0.0	0.0	Winter
17412	2017-01-03 22:00:00	224	5.5	1.5	76.0	23.0	Cloudy	0.0	0.0	Winter
17413	2017-01-03 23:00:00	139	5.0	1.0	76.0	22.0	Scattered Clouds	0.0	0.0	Winter

17414 rows × 10 columns

In [20]: df.to_excel('london_bikes_final.xlsx', index=False, sheet_name='Data')

In []:

